1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC tumors with apparent immune infiltration (TILs) or otherwise favorable classification still recur or progress; why does immune presence fail to translate into durable tumor control in a subset of cases? 2) Tier 1 paths used: I built the core mechanistic hypothesis on MSH6 (MMR → neoantigen load → immune recruitment) and E-FABP (tumor lipid-handling/metabolic programs linked to metastatic/aggressive behavior). These two provide opposing influences on immune efficacy—MSH6 can increase immunogenicity and TIL recruitment, whereas E-FABP-driven metabolism can create a metabolically suppressive microenvironment that impairs T cell function. 3) Tier 2 support: urine-DNA methylation biomarkers offer a noninvasive readout of tumor epigenetic state that can contextualize immune/metabolic phenotypes; HDL-C/CDKN2A link systemic lipid state and cell-cycle control as potential modifiers of tumor aggressiveness; HRR (inflammation/nutrition index) can alter host immune competence; AI-based integrative methods can discover interactions among these layers. 4) Endpoint choice rationale: Progression-free interval (PFI) is the most clinically relevant outcome for NMIBC (recurrence/progression risk and timing drive management), and captures failures of local immune control. 5) Deliberate exclusions: I excluded PSA, SARIFA, and cross-cancer grading analogies from the core hypothesis because they are cross-contextual or less specific to NMIBC biology; I also omitted TLS-like aggregates and TSR as the imaging feature because TIL density maps more directly onto the MSH6 → neoantigen axis and is measurable on H&E for NMIBC. I kept the hypothesis tightly focused (1 imaging feature + 2 anchors); other entities were placed as secondary modifiers for follow-up testing?

Working Hypothesis

In NMIBC, the protective association between high Tumor-Infiltrating Lymphocyte (TIL) density and longer progression-free interval is contingent on MSH6-driven neoantigen generation, but E-FABP–driven lipid metabolic reprogramming in tumor cells creates a metabolically suppressive microenvironment that impairs T cell effector function. Consequently, NMIBC tumors with high TIL density and elevated E-FABP expression will exhibit shorter PFI than tumors with low E-FABP, irrespective of MSH6 status, indicating that metabolic state can override neoantigen-driven immune benefit; urine-DNA methylation biomarkers and other modifiers (HDL-C, CDKN2A, HRR) can refine risk stratification as secondary modifiers.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, E-FABP
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 can increase immunogenicity and TIL recruitment, whereas E-FABP-driven metabolism can create a metabolically suppressive microenvironment that impairs T cell function.

E-FABP: E-FABP-driven metabolism can create a metabolically suppressive microenvironment that impairs T cell function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	CDKN2A	Supporting modifier from knowledge graph
Secondary	HRR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: mediates lipid uptake and fatty acid metabolism; upregulated lipid metabolism in tumor cells and the tumor microenvironment can suppress T cell function through mechanisms such as altered fuel availability, lipid-induced exhaustion, and changes in antigen presentation.

Prediction 2: Retrospective NMIBC cohorts with matched tissue and urine samples to assess MSH6 status, E-FABP expression, and TIL density, plus urine DNM biomarkers;

Prediction 3: Prospective validation in an independent cohort;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- urine-DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.